

UNIVERSITY OF NAMIBIA
DEPARTMENT OF MECHANICAL AND METALLURGICAL ENGINEERING
TEST 2: WEDNESDAY 21st MAY 2025
13641NM: ENGINEERING MECHANICS II
MARKING SCHEME

Question One

Q.1 → 50/50

(a) Given

A wheel with an angular acceleration of $\alpha = (1.5\theta) \text{ rad/s}^2$, where θ is in radians.

Required

To determine the magnitude of the velocity and acceleration of a point P located on the edge after the pulley has rotated 2 revolutions. The pulley has a radius of **0.2 m** and starts from rest.

Solution

At rest, $t = 0$, $\theta_0 = 0$, $\omega_0 = 0$

The angular displacement of the wheel after 2 revolution is given as:

$$\theta = 2 \text{ rev} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} = 12.57 \text{ rad} \quad \checkmark \text{ (2)}$$

Hence, the angular acceleration of the wheel after 2 rev is obtained as

$$\alpha = (1.5\theta) = 1.5 \times 12.57 = 18.855 \text{ rad/s}^2 \quad \checkmark \text{ (2)}$$

$$\text{From } \omega = \frac{d\theta}{dt} \text{ and } \alpha = \frac{d\omega}{dt} \quad \checkmark \text{ (1)}$$

$$\text{i.e. } dt = \frac{d\omega}{\alpha} = \frac{d\theta}{\omega} \quad \checkmark \text{ (1)}$$

$$\omega d\omega = \alpha d\theta \quad \checkmark \text{ (2)}$$

$$\int_{\omega_0}^{\omega} \omega d\omega = \int_{\theta_0}^{\theta} \alpha d\theta = \int_{\theta_0}^{\theta} (1.5\theta) d\theta \quad \checkmark \text{ (1)}$$

$$\therefore \frac{\omega^2}{2} \Big|_0^{\omega} = 1.5 \times \frac{\theta^2}{2} \Big|_0^{\theta}$$

$$\text{or } \omega^2 = 1.5\theta^2 \quad \checkmark \text{ (1)}$$

$$\text{i.e. } \omega = \sqrt{1.5} \times (\theta) \quad \checkmark \text{ (1)}$$

hence, when $\theta = 12.57 \text{ rad}$

$$\omega = \sqrt{1.5} \times (12.57) = 15.4 \text{ rad/s} \quad \checkmark \text{ (2)}$$

The tangential velocity of P is given as:

$$v_p = \omega_p r_p, \text{ where, } r_p = 0.2 \text{ m}$$

$$\therefore v_p = 15.4 \times 0.2 = 3.08 \text{ m/s}$$

$$\text{Ans: } v_p = 3.08 \text{ m/s}$$

Now, the tangential and normal accelerations at P can be obtained as:

$$a_t = \alpha r$$

$$\therefore (a_t)_p = \alpha_p r_p = 18.855 \times 0.2 = 3.771 \text{ m/s}^2$$

$$a_n = \omega^2 r$$

$$\therefore (a_n)_p = \omega_p^2 r_p = (15.4)^2 \times 0.2 = 47.432 \text{ m/s}^2$$

$$\therefore a_p = \sqrt{[(a_t)_p]^2 + [(a_n)_p]^2} = \sqrt{[(3.771)^2 + (47.432)^2]} = 47.58 \text{ m/s}^2$$

$$\text{Ans: } a_p = 47.58 \text{ m/s}^2$$

(b) Given

That the wheel in Q. 3 (a) drives a wheel of **0.5 m** diameter.

Required

- To determine the speed of the wheel Q in revolution per minute (rpm) and
- To determine the acceleration at the outermost part of the pulley Q .

Solution

(i) The tangential velocity of the wheel in Q. 1 (a) and the outer surface of the pulley must be the same. Hence,

$$v_p = \omega_p r_p = v_q = \omega_q r_q = 3.08 \text{ m/s}$$

$$\text{but } r_q = \frac{0.5}{2} \text{ m} = 0.25 \text{ m}$$

$$\therefore \omega_q = \frac{v_q}{r_q} = \frac{3.08}{0.25} = 12.32 \text{ rad/s}$$

$$\text{but } \omega_q = \frac{2\pi N_q}{60}$$

$$\therefore N_q = \frac{60 \times \omega_q}{2\pi} = \frac{60 \times 12.32}{2\pi} = 117.65 \text{ rpm}$$

$$\text{Ans: } N_q = 117.65 \text{ rpm}$$

(ii) To determine the acceleration at a point m , note that m lies at the outermost part of the pulley Q . Hence,

$$r_m = 0.25 \text{ m}$$

$$\therefore (a_t)_m = \alpha_m r_m = \alpha_q r_q = (a_t)_q$$

$$\text{Note that, } \therefore (a_t)_m = (a_t)_q = (a_t)_p = 3.771 \text{ m/s}^2$$

$$\text{Hence, } (a_n)_m = \omega_m^2 r_m$$

$$\text{Where, } \omega_m = \omega_q = 12.32 \text{ rad/s}$$

$$\therefore (a_n)_m = (12.32)^2 \times 0.25 = 37.95 \text{ m/s}^2$$

$$\therefore a_m = \sqrt{[(a_t)_m]^2 + [(a_n)_m]^2} = \sqrt{[(3.771)^2 + (37.95)^2]} = 38.13 \text{ m/s}^2$$

$$\underline{\text{Ans: } a_m = 38.13 \text{ m/s}^2}$$

Q. 2 → 50/50

Question Two

(a) Given

That a planar kinetic analysis of a machine was conducted. And it was required to understand the effect the machine elements has in resisting angular acceleration.

Required

To determine the moment of inertia about the axial axis of the cylinder shown in Figure Q. 2 (a).

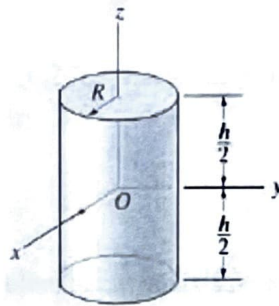
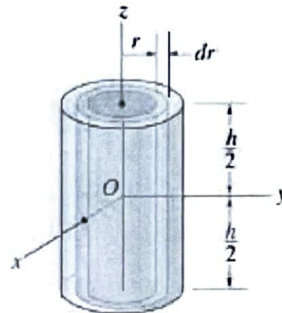


Figure Q. 2 (a)

Solution

Shell Element: This problem can be solved using the shell element as follows:



The volume of the element is $dV = (2\pi r)(dr)(h)$ ✓ ① (1)

so that its mass is

$dm = \rho dV = dm = \rho(2\pi hr dr)$ ✓ ① (2)

Since the *entire element* lies at the same distance r from the z axis, the moment of inertia of the element is

$dI_z = r^2 dm = \rho 2\pi hr^3 dr$ ✓ ② (3)

Integrating over the entire region of the cylinder yields:

$I_z = \int_m r^2 dm = \rho 2\pi h \int_0^R r^3 dr = \frac{\rho\pi}{2} R^4 h$ ✓ ② (4)

But the mass of the cylinder is:

$$m = \int_m dm = \rho 2\pi h \int_0^R r dr = \rho \pi h R^2 \quad \checkmark \textcircled{2} \quad (5)$$

Hence,

$$I_z = \frac{1}{2} m R^2 \quad \checkmark \textcircled{2} \quad (6)$$

(b) Given

That the pendulum shown in Figure Q. 2 (b) is suspended from the pin at O and consists of two thin rods, each having a weight of 5 kg .

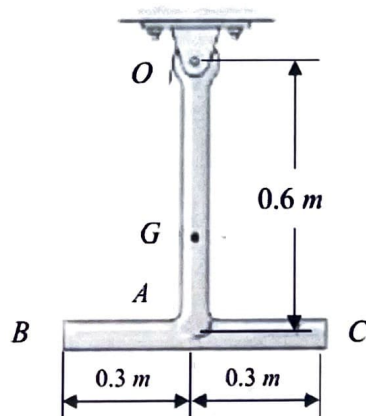
Required

To determine:

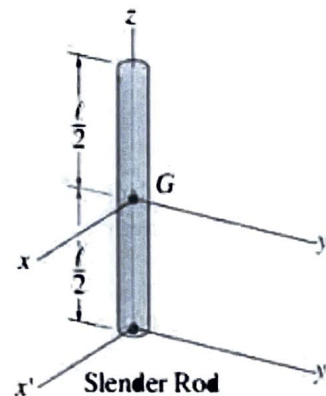
- (i) the moment of inertia of the pendulum about an axis passing through point O ,
- (ii) the mass center G of the pendulum relative to point O and
- (iii) the moment of inertia about G .

The moment of inertia of a slender rod shown in Figure Q. 2(c) is given as:

$$I_{xx} = I_{yy} = \frac{1}{12} m l^2; I_{x'x'} = I_{y'y'} = \frac{1}{3} m l^2; I_{z'z'} = 0$$



(b)



(c)

Figure Q. 2

Solution

- (i) The moment of inertia of the rod OA about an axis perpendicular to the page and passing through point O of the rod is given as:

$$(I_{OA})_O = \frac{1}{3} m l^2 = \frac{1}{3} \times 5 \times (0.6)^2 = 0.6 \text{ kgm}^2 \quad \checkmark \textcircled{4}$$

For rod BC we have:

$$(I_{BC})_O = \frac{1}{12}ml^2 + md^2 = \frac{1}{12} \times 5 \times (0.6)^2 + 5 \times (0.6)^2 = 0.15 + 1.8 \\ = 1.95 \text{ kgm}^2$$

The moment of inertia of the pendulum about O is therefore:

$$I_O = 0.6 + 1.95 = 2.55$$

Ans: $I_O = 2.55 \text{ kgm}^2$

(ii) The mass center G relative to point O, will be located by OG, or $\bar{y}$. This can be expressed as:

$$\bar{y} = \frac{\sum \bar{y}m}{\sum m} = \frac{\bar{y}_1 m_1 + \bar{y}_2 m_2}{m_1 + m_2} = \frac{0.3 \times 5 + 0.6 \times 5}{5 + 5} = \frac{4.5}{10} = 0.45 \text{ m}$$

Ans: $\bar{y} = 0.45 \text{ m}$

(iii) The moment of inertia I_G may be determined as follows:

$$I_O = I_G + md^2$$

$$\text{i.e., } I_G = I_O - md^2$$

$$\text{but, } I_O = 2.55 \text{ kgm}^2, m = 5 + 5 = 10 \text{ kg}, d = \bar{y} = 0.45 \text{ m}$$

$$I_G = 2.55 - 10 \times (0.45)^2 = 2.55 - 2.025 = 0.525$$

Ans: $I_G = 0.525 \text{ kgm}^2$

Q. 3 → 50/50

Question Three

(a) Given

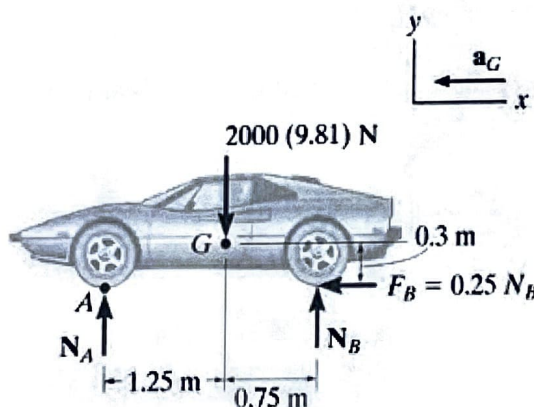
A car A which has a mass of $2,000 \text{ kg}$ and a center of mass at G . Neglect the mass of the wheels. The coefficient of kinetic friction between the wheels and the road is $\mu_k = 0.25$. Take the distance of the front and rear wheels from G to be 1.25 m and 0.75 m , while the height of G is 0.3 m respectively.

Required

To determine (i) the acceleration, (ii) the reactions at the front and rear wheels, if the rear "driving" wheels are always slipping, whereas the front wheels are free to rotate.

Solution

The free body diagram of the car A is shown as follows:



Equations of motion:

$$(+\rightarrow) \sum F_x = m(a_G)_x \quad \checkmark \text{ (3)} \quad (1)$$

$$\text{i.e., } -0.25N_B = -(2000 \text{ kg})a_G \quad \checkmark \text{ (3)} \quad (2)$$

$$+\uparrow \sum F_y = m(a_G)_y \quad \checkmark \text{ (3)} \quad (3)$$

$$\text{i.e., } N_A + N_B - 2000 \times 9.81 \text{ N} = 0 \quad \checkmark \text{ (3)} \quad (4)$$

$$+\curvearrowright \sum M_G = 0 \quad \checkmark \text{ (2)} \quad (5)$$

$$\text{i.e., } -N_A(1.25 \text{ m}) - 0.25N_B(0.3 \text{ m}) + N_B(0.75 \text{ m}) = 0 \quad \checkmark \text{ (3)} \quad (6)$$

Solving Eqs. (2), (4) and (6) gives:

Ans: $a_G = 1.59 \text{ m/s}^2 \leftarrow$ ✓ (3)

$N_A = 6.88 \text{ kN}$ ✓ (3)

$N_B = 12.7 \text{ kN}$ ✓ (3)

(b) Given

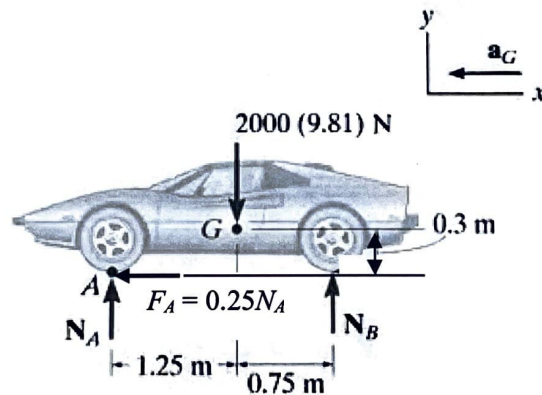
Assuming a different car B was designed to function conversely to the car A in Question 3 (a). If the front "driving" wheels of car B are always slipping, whereas the rear wheels are free to rotate.

Required

To determine (i) the acceleration, (ii) the reactions at the front and rear wheels, assuming all other specifications of car B are similar to those of car A .

Solution

The free body diagram of the car A is shown as follows:



Equations of motion:

$(+\rightarrow) \sum F_x = m(a_G)_x$ ✓ (2) (3)

i.e., $-0.25N_A = -(2000 \text{ kg})a_G$ ✓ (3)

$+\uparrow \sum F_y = m(a_G)_y$ ✓ (2)

i.e., $N_A + N_B - 2000 \times 9.81 \text{ N} = 0$ ✓ (3)

$+\curvearrowright \sum M_G = 0$ ✓ (2)

i.e., $-N_A(1.25 \text{ m}) - 0.25N_A(0.3 \text{ m}) + N_B(0.75 \text{ m}) = 0$ ✓ (3)

Solving Eqs. (2), (4) and (6) gives:

Ans: $a_G = 0.8863 \text{ m/s}^2 \leftarrow$ ✓ (3)

$N_A = 7.09 \text{ kN}$ ✓ (3)

$N_B = 12.53 \text{ kN}$ ✓ (3)